

1	2	3	Σ
2/4	1.5/2	3/8	6.5/14

CPS I — Exercise Sheet 8

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Exercise 1: Invariant checking I

2/4

$U = \emptyset, \pi = \emptyset$ ✓

- $\text{DFS}(S_0, a) U = \{S_0\}, \pi = \{S_0\}$ ✓
 - $\text{DFS}(S_3, a) U = \{S_0, S_3\}, \pi = \{S_0, S_3\}$ ✓
 - $\text{DFS}(S_3, a) U = \{S_0, S_3\}, \pi = \{S_0, S_3, S_3\}$ 2 < 3, so we would call $\text{DFS}(S_2, a)$ first.
 - $\text{DFS}(S_2, a) U = \{S_0, S_3, S_2\}, \pi = \{S_0, S_3, S_2\}$
 - $\text{DFS}(S_0, a) U = \{S_0, S_3, S_2\}, \pi = \{S_0, S_3, S_2, S_0\}$ here, we would pop S_0 since we already visited it. Then we would pop S_2 , since we called $\text{DFS}(S_0)$ in the call $\text{DFS}(S_2)$. -0.5
- $\text{DFS}(S_1, a) U = \{S_1\}, \pi = \{S_1\}$ ($U = \{S_0, S_2, S_3, S_4\}$ (we never reset U .) -0.5)
 - $\text{DFS}(S_3, a) U = \{S_1, S_3\}, \pi = \{S_1, S_3\}$ we would pop S_3 because we visited it already -0.5
 - ~~$\text{DFS}(S_3, a) U = \{S_1, S_3\}, \pi = \{S_1, S_3, S_3\}$~~ we can then do $\text{DFS}(S_4)$
 - $\text{DFS}(S_4, a) U = \{S_1, S_4\}, \pi = \{S_1, S_4\}$ ✓
 - ~~$\text{DFS}(S_3, a) U = \{S_1, S_4, S_3\}, \pi = \{S_1, S_4, S_3\}$~~
 - ~~$\text{DFS}(S_3, a) U = \{S_1, S_4, S_3\}, \pi = \{S_1, S_4, S_3, S_3\}$~~ we stop once we found a violation -0.5

$S_0, S_1, S_2, S_3 \models a$

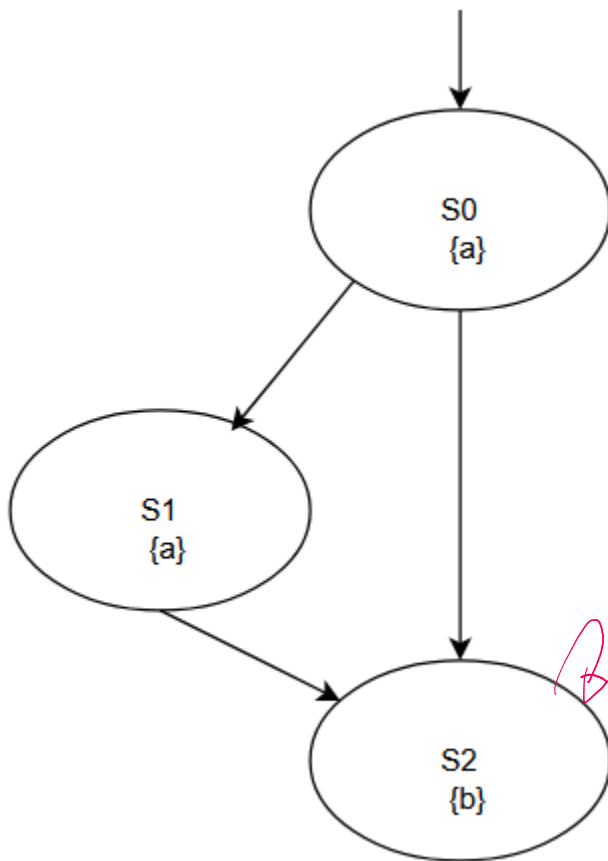
$S_4 \not\models a$

$S_0 \not\models$ always a counter example: S_1, S_4

✓

Exercise 2: Invariant checking II

1.5/2



In the lecture, we have exclusively
considered TS without terminal states
for a few weeks now.

The satisfaction relation for LT
properties, including invariants, has
only been defined for such TS as
well. So I have to
do -0.5

$$\Phi = a$$

The counterexample with non-minimal length S0,S1,S2

A counterexample of minimal length S0,S2

Exercise 3: Invariants

3/8

(a) No Idea

(b)

- $\sigma = (\{a\}\emptyset)^\omega$
 - For $A_i = \{a\}$, $\sigma = (\{a\}\{b\})^\omega$, $\sigma \in E$ ✓
 - For $A_i = \emptyset$, $\sigma = (\emptyset)^\omega$, $\sigma \in E$ ✓
- $\sigma = (\{a\}\{b\})^\omega$
 - For $A_i = \{a\}$, $\sigma = \{a\}^\omega$, $\sigma \in E$ ✓
 - For $A_i = \{b\}$, $\sigma = \{b\}^\omega$, $\sigma \in E$ ✓

0/5

3/3

- $\sigma = (\{a\}\emptyset)^\omega$ no commas
 - For $A_i = \{a\}$, $\sigma = \{a\}\{a\}\emptyset^\omega$, $\sigma \in E$ ✓
 - For $A_i = \emptyset$, $\sigma = \{a\}\{a\}\emptyset^\omega$, $\sigma \in E$ ✓